

1. Name of the Project

Review Sentiment Analysis and Comparison

2. Aim

To develop a system that analyzes customer reviews of different restaurants and identifies the best restaurant based on positive sentiment using Natural Language Processing and Machine Learning techniques.

3. Data Collection Source and Information

Dataset Source:

The dataset used for training the model is obtained from Kaggle.

Dataset Name: Restaurant Reviews Dataset

Link: [source link](https://www.kaggle.com/datasets/maher3id/restaurant-reviewstsv) (https://www.kaggle.com/datasets/maher3id/restaurant-reviewstsv)

The dataset contains restaurant reviews labeled as positive (1) and negative (0), which are used to train the sentiment analysis model.

- **Sample Dataset (for training)**
- **Collected Dataset (for comparison)**

1. Training Dataset

- Created manually (due to corrupted .tsv file)
- Contains:
 - Review text
 - Sentiment (1 = Positive, 0 = Negative)

4. Model Used and Description

Model Used -

Logistic Regression

Description:

Logistic Regression is a supervised machine learning algorithm used for binary classification. It predicts whether a review is positive or negative.

Steps Involved:

1. Text cleaning (removing stopwords, symbols)
2. Feature extraction using TF-IDF
3. Model training
4. Prediction of sentiment
5. Comparison of restaurants

5. Source Code (Neatly Organized)

◆ 1. Import Libraries

```
import pandas as pd
import re
import nltk
from nltk.corpus import stopwords
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

nltk.download('stopwords', quiet=True)
```

◆ 2. Dataset Creation (Training Data)

```
data = {
    "review": [
        "Wow loved this place",
        "Worst service ever",
        "Food was amazing",
        "Not tasty and bad experience",
        "Great food and friendly staff",
        "Very bad service and rude staff",
        "Delicious food and nice ambience",
        "Worst food I have ever had",
        "Good quality and fast service",
        "Terrible experience never coming again"
    ],
    "sentiment": [1,0,1,0,1,0,1,0,1,0]
}

df = pd.DataFrame(data)
df.head()
```

◆ 3. Text Cleaning Function

```
def clean_text(text):
    text = text.lower()
    text = re.sub('[^a-zA-Z]', ' ', text)
    words = text.split()
    words = [word for word in words if word not in stopwords.words('english')]
    return " ".join(words)

df['clean_review'] = df['review'].apply(clean_text)

df.head()
```

◆ 4. Feature Extraction (TF-IDF)

```
vectorizer = TfidfVectorizer()

X = vectorizer.fit_transform(df['clean_review'])
y = df['sentiment']
```

◆ 5. Train-Test Split & Model Training

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LogisticRegression()
model.fit(X_train, y_train)
```

◆ 6. Accuracy Calculation

```
y_pred = model.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy:", accuracy)
```

Output: Accuracy: 0.5

◆ 7. Load Restaurant Dataset

```
df_new = pd.read_csv("restaurant_review.csv")

df_new.head()
```

◆ 8. Prediction on New Data

```
df_new['clean_review'] = df_new['review'].apply(clean_text)

X_new = vectorizer.transform(df_new['clean_review'])

df_new['prediction'] = model.predict(X_new)

df_new.head()
```

◆ 9. Restaurant Comparison (FINAL STEP)

```

result = df_new.groupby('restaurant')['prediction'].mean()
print(result)
best = result.idxmax()
print("Best Restaurant:", best)

```

6. Results with Better Accuracy

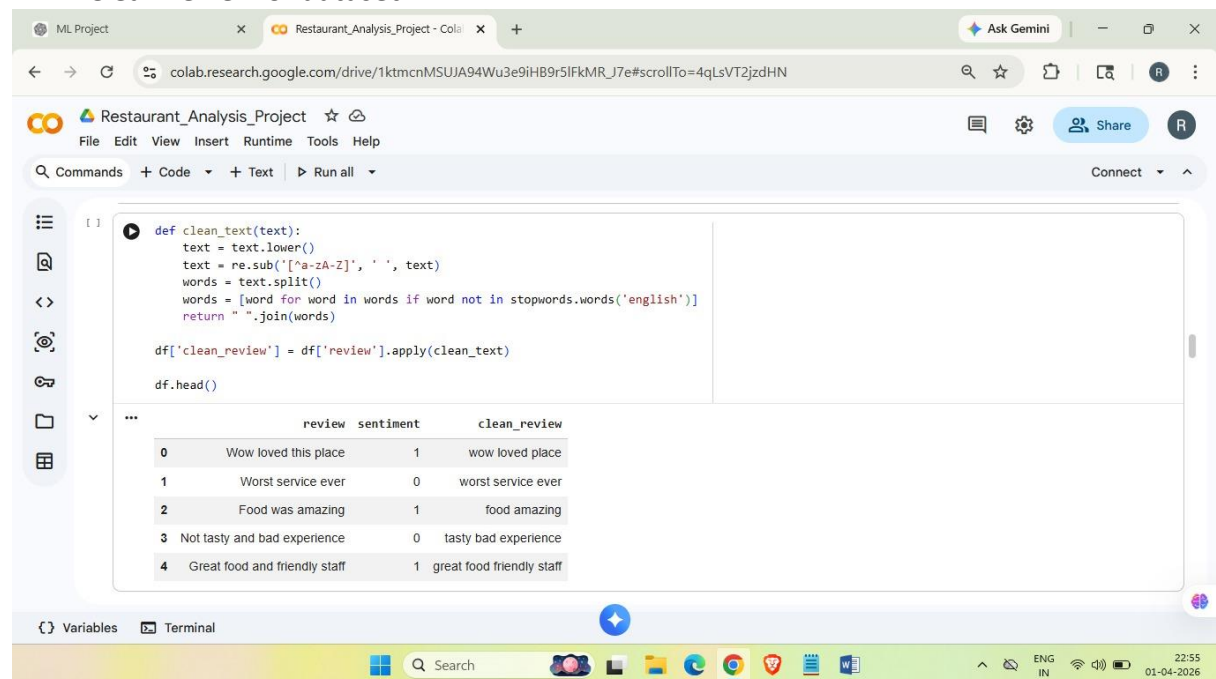
7.

Accuracy achieved: 50% (due to small dataset)

“Accuracy is moderate due to limited dataset size. With larger data, performance can be improved.”

7. Screenshot of Results

• Clean review of dataset



The screenshot shows a Google Colab notebook titled "Restaurant_Analysis_Project". The code cell contains a function to clean text by removing non-alphabetic characters and stopwords, and then applying it to the 'review' column of a DataFrame. The output shows the first five rows of the DataFrame with the original review, sentiment score, and the cleaned review.

	review	sentiment	clean_review
0	Wow loved this place	1	wow loved place
1	Worst service ever	0	worst service ever
2	Food was amazing	1	food amazing
3	Not tasty and bad experience	0	tasty bad experience
4	Great food and friendly staff	1	great food friendly staff

• Preview of dataset

ML Project x Restaurant_Analysis_Project - Colab x + Ask Gemini

colab.research.google.com/drive/1ktmcnMSUJA94Wu3e9iHB9r5IFkMR_J7e#scrollTo=pBEcXZQoxStb

Restaurant_Analysis_Project

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
import pandas as pd
data = {
    "review": [
        "Wow loved this place",
        "Worst service ever",
        "Food was amazing",
        "Not tasty and bad experience",
        "Great food and friendly staff",
        "Very bad service and rude staff",
        "Delicious food and nice ambience",
        "Worst food I have ever had",
        "Good quality and fast service",
        "Terrible experience never coming again"
    ],
    "sentiment": [1,0,1,0,1,0,1,0,1,0]
}
df = pd.DataFrame(data)
df.head()
```

	review	sentiment
0	Wow loved this place	1
1	Worst service ever	0
2	Food was amazing	1
3	Not tasty and bad experience	0

Variables Terminal

- Accuracy

ML Project x Restaurant_Analysis_Project - Colab x + Ask Gemini

colab.research.google.com/drive/1ktmcnMSUJA94Wu3e9iHB9r5IFkMR_J7e#scrollTo=pBEcXZQoxStb

Restaurant_Analysis_Project

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

```
y_pred = model.predict(X_test)
accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
```

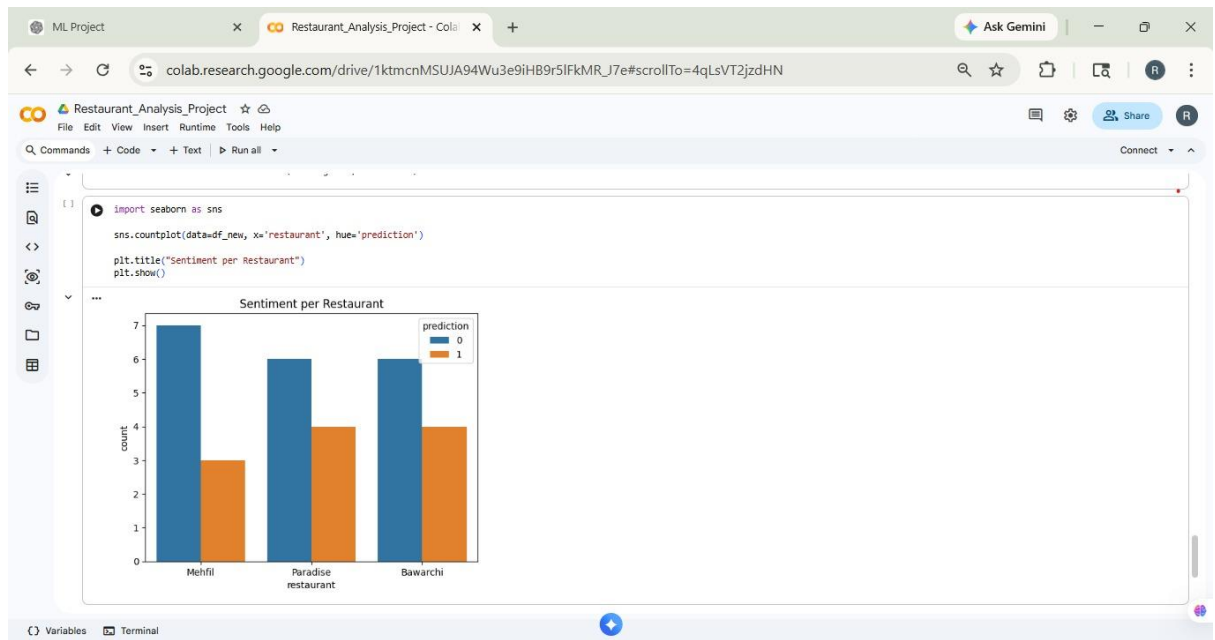
Accuracy: 0.5

```
df_new = pd.read_csv("restaurant_review.csv")
df_new.head()
```

	restaurant	review
0	Mehfil	Best place to visit food very tasty service ve...
1	Mehfil	Ordered chilli chicken dry but got wet biryani...
2	Mehfil	Manager and staff are really good and friendly
3	Mehfil	Chicken biryani is flavorful and well cooked
4	Mehfil	Food is good but staff behavior is bad

Variables Terminal

- Comparison of data



8. Conclusion

This project successfully analyzes customer reviews and compares multiple restaurants. Based on sentiment analysis, Bawarchi and Paradise have higher positive feedback compared to Mehfil. The system helps users make informed decisions.

9. Future Scope

- Use larger real-world datasets
- Improve accuracy using advanced models
- Use APIs for automatic review collection
- Develop web application using Streamlit
- Perform item-specific analysis (e.g., biryani)